

3DP-Ori: Bridging-Printing Based Origami Fabrication Method with Modifiable Haptic properties

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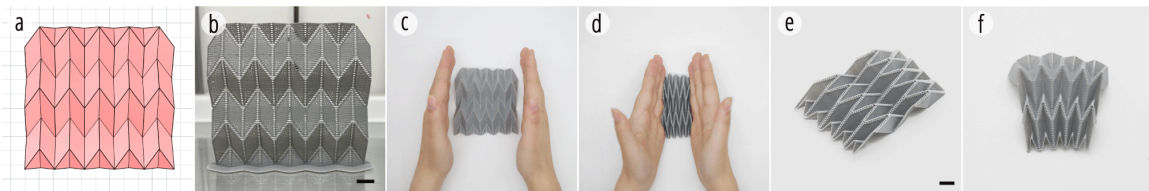


Figure 1: Fabrication process of 3DP-Ori: (a) virtual generated models; (B) printed objects; (C, D) manual deformation; (E, F) object keeps its steady shape as desired.

ABSTRACT

Origami is the art of making 2D/3D shapes by folding paper, HCI researchers have leveraged novel structure design and material techniques to create a wide range of origami-based shape-changing interface. Meanwhile, additive manufacturing which could easily fabricate complicated artifacts and augment the haptic sensation, has drawn more attention to the field. This paper presents 3DP-Ori, a fabrication method of flexible origami construction with conventional FDM printers. To create and end-to-end pipeline, we developed a parametric design tool with dynamic folding simulation

and feasibility estimation, which enable our software to output a printable semi-folded origami model for further easy manual deformation. Adopting and optimizing bridging-based printing, we leverage multiple geometric patterns with adjustable flexibility on creases, which further effect the haptic properties on 3DP-Ori objects. We believe the 3DP-Ori extends the design space of 3D printing beyond typically hard and fixed forms, and it will help designers and researchers to fabricate interactivities with various physical properties.

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CCS CONCEPTS

• **Human computer interaction (HCI)** → Interactive systems and tools.

KEYWORDS

Origami, Shape-changing interfaces, Personal fabrication, 3D printing

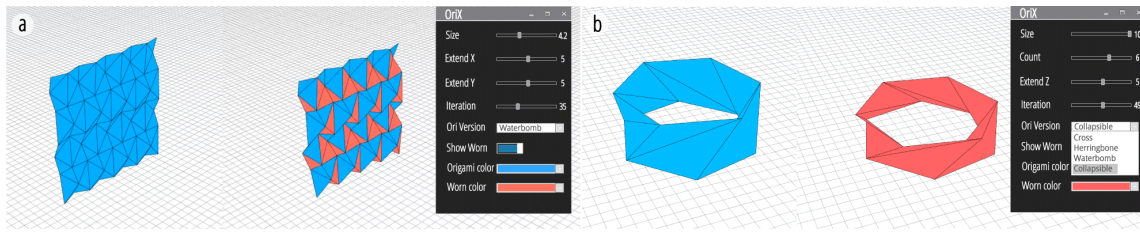


Figure 2: 3DP-Ori interface: simulation and estimation of (a) 6 creased-waterbomber and (b) Collapsible structure.

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1 INTRODUCTION

In the Human-computer Interaction (HCI), origami structure is widely used for shape-changing interface, prototyping and material application [6, 12, 13] since it is flexible [8, 9, 11], light and saves material. Additionally, manufacture objects with flat patterns then deform into 3D structures has also been thoroughly demonstrated in origami related works [1, 3, 5, 10]. For instance, the physical and haptic properties of Origami are revealed and developed by Chang [1], Ion [4] and Hao [3]. As the size of print platform limits the design space of prototyping, researcher developed folded prints which are able to be stretched or unfolded after printed [2, 7]. As a result, a growing interest in developing origami design, fabrication, and modification methods that allow users to receive efficient and easy origami-making is revealed by recent works.

However, folding complicated origami patterns on flat piece requires a relative long time and sophisticated skills as creases are trendless and may conflict each other. In this poster, we will first explain the folding simulation process printable supervisor of 3DP-Ori, followed by the fabrication strategy of design software. We will then leverage two applying scenarios. Our main contributions include:

- semi-folded state of 3DP-Ori objects making manual folding process much more efficient;
- combining the flexible and stable manufacture strategies of single material (creases are bridging-based printed and other parts are conventional printed);
- adjustable parameters of flow multiplier and printing speed creates various physical properties;
- simulating and estimating folding process to output feasible state of several origami structure.

2 COMPUTATIONAL DESIGN

Using Grasshopper, a parametric design plugin for Rhinoceros [14], an 3DP-Ori user interface (Figure 2) is developed to simulate and estimate origami folding process. At present, 5 classic origami types have been created and tested, which allow users to customize folding iteration to control angles of creases. At the same time, 3DP-Ori

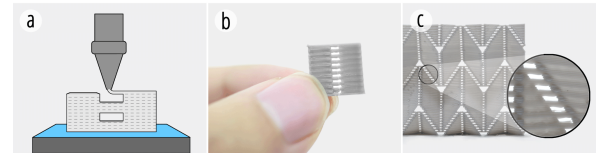


Figure 3: (a) bridging based 3D printing concept (b) tiny sample (c) the creased made with bridging.

supervises and estimates this process, which would mark the parts red (Figure 3) if it is unprintable. Thus, outputted blue virtual models would be feasible for conventional FDM printers. The interface also allows users to change the size of the objects (Extend X and Y refers to count of creases of two directions).

3 MANUFACTURE STRATEGIES

Conventional FDM printing requires solid base or support for layer-by-layer fabrication. However the fused line from printer has a certain toughness, which enable it resist gravity in short distance. Bridging-based printing is used to manufacture overhang polymer without supports, which have already been applied on various inspiring 3DP models. Additionally, our adopt and optimize the toughness of single sheet of this structure, which afford folding repeatedly. Figure 3c shows that we optimized and adopt this technique to fabricate flexible and compliant creases.

4 APPLICATIONS

4.1 Buttons with different haptic properties

Since the physical properties of WireTruss are directly related to the flow multiplier and print speed, we tested to apply this feature on buttons. We completed 4 buttons with 3 layers of collapsible structure with flow 0.8, 1.0, 1.2 and 1.6. The artifacts have same appearance, but perform different resistance when they are pressed.

4.2 Fashion design

Compared with conventional method, the 3DP-Ori workflow not only allow designers to create origami shapes from flat, but also empowers them to fabricate objects with curved surfaces which are connected with origami creases. This feature full-fill the requirement of combining origami texture with complicated surfaces. Figure 6 demonstrate this contribution of 3DP-Ori by showing a fashion design application. The dress is printed within 2 hours and the flexibility of it fit the doll's waist well.

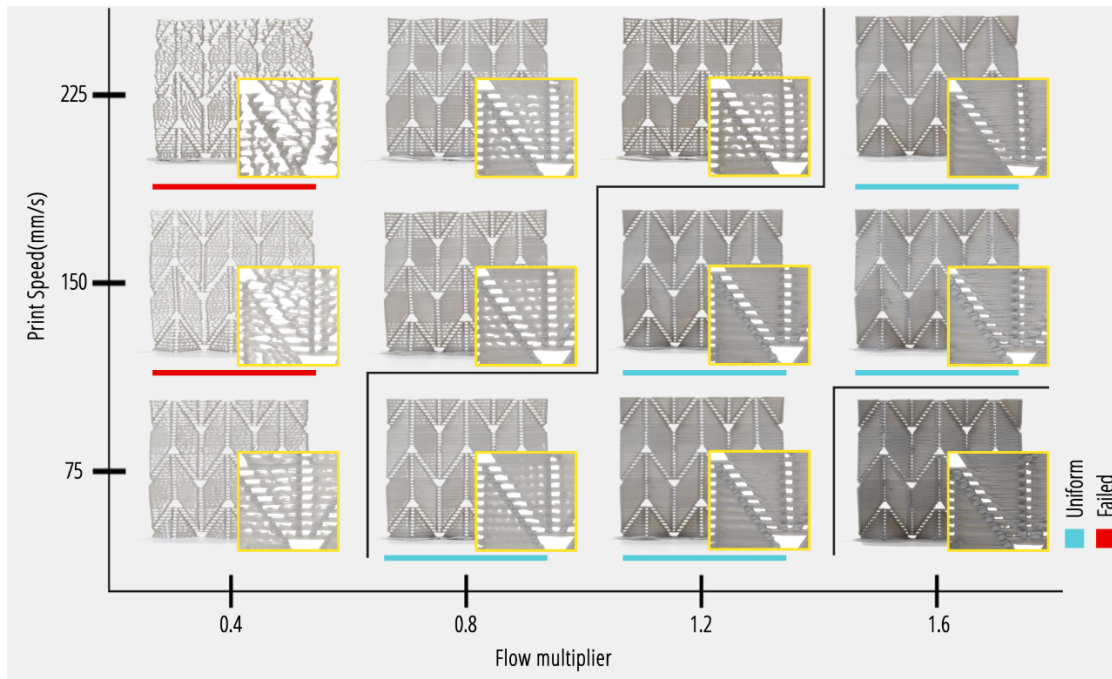


Figure 4: Results of 3DP-Ori objects with different flow multipliers and print speed. The samples with blue underlined could be well folded

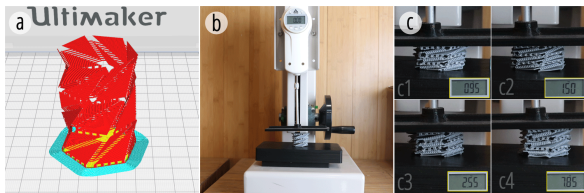


Figure 5: Button prototype made by 3DP-Ori: (a)printing path in Cura; (b) resistance measure; (c) different button made with flow multiplier 80%, 100%, 120%,160% with their measured resistance at right bottom corner



Figure 6: Dress design enabled by 3DP-Ori

5 CONCLUSION

In this work, we have demonstrated that bridging based printing brings the possibility of efficient origami fabrication for makers and designers using consumer-grade 3D printers. We developed a computational design tool to output origami models with flexible creases, which can be directly 3D printed as well as easily folded.

The application examples made by our design tool showcased multi-form, and multi-use applications with shape-changing, physical property-changing features. We look forward to extending our method into future sustainable fabrication and interaction.

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